==
MASTER PREPARATION NOTES – CONCEPTS & GLOSSARY
(Python • SQL • Data Engineering • AI • Analytics)
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1. ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Artificial Intelligence (AI):
Field of computer science focused on building systems that mimic human intelligence such as reasoning, learning, and decision-making.

Machine Learning (ML):
A subset of AI where systems learn patterns from data instead of being explicitly programmed.

Deep Learning (DL):
A subset of ML using neural networks with multiple layers, mainly for image, speech, and language tasks.

Supervised Learning:
ML technique where models learn from labeled data (input + known output).

Unsupervised Learning:
ML technique that finds hidden patterns in unlabeled data without predefined outcomes.

Reinforcement Learning:
Learning approach where an agent improves behavior through rewards and penalties from an environment.

Artificial General Intelligence (AGI):
Theoretical AI capable of human-level intelligence across multiple tasks; does not exist today.

AI Limitation:
Modern AI systems can generate incorrect, biased, or hallucinated outputs due to probabilistic nature.

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2. PROMPT ENGINEERING & GENERATIVE AI

Prompt Engineering:
The practice of designing precise and structured inputs to guide AI model outputs effectively.

Large Language Model (LLM):
A model trained on massive text data to predict and generate human-like language.

Generative AI:

AI systems that generate new content such as text, code, images, or summaries.

Key Reality:
Generative models do not "understand" content; they predict statistically likely responses.

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3. DATA WAREHOUSING CONCEPTS

Data Warehouse:
Centralized repository that stores integrated, historical data for analytics and reporting.

Staging Layer:
Temporary storage area for raw data extracted from source systems before processing.

Integration Layer:
Layer where data is cleaned, standardized, joined, and transformed.

Access Layer:
Layer optimized for end users, reports, dashboards, and data marts.

Enterprise Data Warehouse (EDW):
Organization-wide data repository serving multiple business domains.

Data Mart:
Subject-specific subset of data tailored for a department or use case.

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4. DIMENSIONAL MODELING

Fact Table:
Stores measurable, numeric business metrics such as sales, revenue, or quantity.

Dimension Table:
Stores descriptive attributes that provide context to facts (customer, product, date).

Grain:
The level of detail represented by a fact table.

Slowly Changing Dimension (SCD):
Technique for managing changes in dimension data over time.

SCD Type 1:
Overwrite old data; no history maintained.

SCD Type 2:
Maintain full history using multiple rows with effective dates.

SCD Type 3:

Maintain limited history using additional columns.

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5. PYTHON CORE CONCEPTS

List:
Mutable, ordered collection that allows duplicate values.

Tuple:
Immutable, ordered collection; cannot be modified after creation.

Set:
Unordered collection that stores unique elements only.

Dictionary:
Key-value data structure optimized for fast lookups.

Mutability:
Ability of an object to be modified after creation.

break:
Terminates loop execution immediately.

continue:
Skips current loop iteration and moves to the next one.

pass:
Placeholder statement that performs no operation.

Function:
Reusable block of code defined using the def keyword.

Default Argument:
Parameter value used when no argument is provided.

Lambda Function:
Anonymous one-line function used for short operations.

Return Value:
Output sent back by a function; absence of return yields None.

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6. NUMPY FUNDAMENTALS

NumPy:
Library for high-performance numerical computing in Python.

ndarray:

Core NumPy data structure representing a homogeneous array.

Vectorization:
Applying operations on entire arrays without explicit loops.

Broadcasting:
Mechanism allowing NumPy to perform operations on arrays of different shapes efficiently.

Element-wise Operation:
Operation applied independently to each array element.

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7. PANDAS DATAFRAME OPERATIONS

DataFrame:
Two-dimensional labeled data structure similar to a table.

Series:
One-dimensional labeled array.

.loc:
Label-based selection of rows and columns.

.iloc:
Position-based selection of rows and columns.

groupby:
Operation that splits data, applies aggregation, and combines results.

Aggregation:
Computing summary values such as sum, mean, min, max.

replace:
Method used to substitute specific values.

fillna:
Method used to replace missing values.

astype:
Method used to change data types of columns.

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8. DATA VISUALIZATION

Matplotlib:
Low-level plotting library for creating static visualizations.

Seaborn:

High-level visualization library built on Matplotlib for statistical plots.

Scatter Plot:
Shows relationship between two numeric variables.

Pair Plot:
Grid of plots showing pairwise relationships between variables.

Style:
Predefined aesthetic configuration applied to plots.

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9. TESTING FRAMEWORKS

Unit Testing:
Process of testing individual components of code.

unittest:
Built-in Python testing framework using test classes.

TestCase:
Base class for defining test methods in unittest.

pytest:
Third-party testing framework with simple syntax and conventions.

Test Discovery:
Automatic detection of test functions or files.

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10. FILE HANDLING & JSON

File Mode:
Defines how a file is opened (read, write, append, create).

Write Mode (w):
Creates or overwrites a file.

Append Mode (a):
Adds content to an existing file.

JSON:
Lightweight data-interchange format using key-value pairs.

json Module:
Python module used to read and write JSON files.

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11. SQLITE & SQLALCHEMY

SQLite:
Lightweight, file-based relational database.

Cursor:
Object used to execute SQL statements and fetch results.

SQLAlchemy:
Python ORM and SQL toolkit.

ORM (Object Relational Mapping):
Maps database tables to Python classes.

Engine:
SQLAlchemy component managing database connections.

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12. SQL FUNDAMENTALS

DDL:
SQL commands that define database structure (CREATE, DROP).

DML:
SQL commands that manipulate data (INSERT, UPDATE, DELETE).

SELECT:
Retrieves data from tables.

INSERT:
Adds new rows to a table.

UPDATE:
Modifies existing rows.

DELETE:
Removes selected rows.

TRUNCATE:
Removes all rows quickly without row-level logging.

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13. SQL VIEWS & STORED LOGIC

View:
Virtual table defined by a SELECT query.

Updatable View:

View that allows INSERT, UPDATE, DELETE under specific conditions.

Stored Procedure:
Precompiled SQL logic stored in the database.

Benefit:
Improves reusability, security, and performance consistency.

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14. AGGREGATION & FILTERING

GROUP BY:
Groups rows for aggregation.

WHERE:
Filters rows before aggregation.

HAVING:
Filters groups after aggregation.

Aggregate Function:
Function that computes summary values (SUM, AVG, COUNT).

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15. SUBQUERIES & CTEs

Subquery:
Query nested inside another SQL query.

Correlated Subquery:
Subquery that depends on outer query values.

Derived Table:
Subquery used in the FROM clause.

CTE (Common Table Expression):
Temporary named result set defined using WITH.

Multiple CTEs:
Multiple logical steps defined in a single WITH clause.

Recursive CTE:
CTE that references itself for hierarchical or sequence data.

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16. WINDOW FUNCTIONS

Window Function:

Performs calculations across a set of related rows.

ROW_NUMBER:
Assigns unique sequential numbers.

RANK:
Assigns rank with gaps for ties.

DENSE_RANK:
Assigns rank without gaps.

NTILE:
Divides rows into equal groups.

LAG:
Accesses previous row value.

LEAD:
Accesses next row value.

PARTITION BY:
Defines independent groups for window calculations.

ORDER BY (Window):
Controls calculation order inside the window.

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17. TRANSACTIONS & SECURITY

Transaction:
Logical unit of work in a database.

COMMIT:
Permanently saves changes.

ROLLBACK:
Reverts uncommitted changes.

SAVEPOINT:
Marks a point within a transaction.

GRANT:
Assigns permissions to users or roles.

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18. AUTOMATION CONCEPTS

Automation Script:

Program designed to execute repetitive tasks automatically.

Use Case:
Data loading, validation, reporting, monitoring.

Key Benefit:
Consistency, efficiency, and reduced manual errors.

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END OF MASTER PREPARATION NOTES
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